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①

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17/03/25

Q1] What is AI? Considering COVID-19 pandemic situation, how AI helped to survive & renovated our way of life with different applications?

Ans ① Artificial Intelligence refers simulation of human intelligence in machines that are programmed to think & learn like humans
② It enables computers to perform tasks that typically require human intelligence

Applications →

a) Disease Detection & Diagnosis: Analyze CT scan & X-rays to detect covid with high accuracy when testing kits were scarce

b) Drug Discovery & Vaccine Development: Identify potential drug candidate & vaccine formulation by analyzing molecular structures & predicting protein interactions

c) Contact Tracing & monitoring → Track infection spread & identify hotspots areas

d) Healthcare resource allocation → forecast patient surges & optimize resource allocation like ICU beds, ventilators & medical staff

e) Telemedicine: AI powered diagnostic tools & chatbots provided preliminary medical consultations

Q2] Explain agent terminologies

Ans ① Agent → An entity that perceives its environment through sensors & acts upon it through actuators to achieve specific goals

Eg:- Robot Vacuum cleaner perceives dust & obstacles through sensors & navigates around cleaning floors

2) Environment $\rightarrow$ The external surroundings in which an agent operates

Eg:- Roads, traffic signals, pedestrians for a self-driving car

3) Percept $\rightarrow$ Information received by the agent from the environment through its sensors

Eg:- AI receives current board position as percept for chess

4) Percept Sequence $\rightarrow$ The complete history of percepts an agent has received

5) Agent function $\rightarrow$ A mapping from percept sequences to actions.

Eg:- A function that maps a sequence of observed chess positions to next best move

6) Performance Measure $\rightarrow$ A criteria used to evaluate the success of an agent's behavior.

Eg:- For recommendation system, the percentage of suggested items that users actually purchase

7) Rationality $\rightarrow$ Ability of agent to select actions that maximize its performance measure based on its percept sequence & builtin knowledge

Eg:- Shortest path for navigation in traffic

8) Autonomy $\rightarrow$ Ability of agent to operate independently without human intervention

Eg:- A mass over performing navigation without human control

9) Knowledge base $\rightarrow$ A collection of facts, rules, & information that agent use for reasoning

10) Actuators & Sensors $\rightarrow$ Input & output component that allow agent to interact with environment.

Q3] How AI is used to solve 8 puzzle technique?

Ans It involves a 3×3 grid with 8 numbered tiles & an empty space. The goal is to rearrange the tiles from initial to goal state by sliding tiles into empty space

① Search: State-Space representation $\rightarrow$

Each state represents a particular arrangement of tiles on board. An initial & goal state & actions are move (up, down, left, right) that shift a tile into empty space.

② Search Algo $\rightarrow$

a) Uninformed Search:

BFS $\rightarrow$ Explore all possible states at the depth before moving to next depth level

b) DFS $\rightarrow$ Explore as far as possible along a branch before backtracking

c) IDS $\rightarrow$ Combines completeness of BFS with space efficiency of DFS

b) Informed Search :- Heuristic Search

i) Hamming distance (h_1) $\rightarrow$ no. of misplaced tiles compared to goal state

ii) Manhattan Distance (h_2) $\rightarrow$ Sum of horizontal & vertical distances each tile is from its goal position

$$d = \sum |x - x_1| + |y - y_1|$$

2	8	3
1	6	4
7		5

initial



1	2	3
8		4
7	6	5

goal

$$h_1 = 4$$

$$h_2 = 2 + 1 + 1 + 1 = 5$$

Time & Space Complexity = $O(b^d)$

Q4 State PEAS for the following :-
 Ans PEAS descriptor framework is used for describing & analyzing AI agents by specifying these 4 essential elements

Agent	Performance	Environment	Actuators	Sensors
Taxi Driver	Safe arrival, min time/distance, comfort, follow traffic rule	Roads, traffic, passengers, weather conditions	Steering, wheel, brake, turn, horn, accelerator	Camera, distance, Sensors, GPS, fuel gauge
medical Diagnosis system	accuracy, speed, minimizing no. of unnecessary tests	patient with symptoms, medical history db, disease KB	Treatment & test recommend, ations, display interface	Image Scanners (X-ray), patient symptom i/p system, medical test/f
music Composer	Originality, quality, specific genre/style, emotional impact	musical traditions, current trends, preferences	note/rhythm generation, instruments	previously composed DB, style templates, a creative analysis
aircraft autoland	safely landing, comfort, fuel efficiency	runway, air traffic & state	control surface, gears, engine	altimeter, air speed indicator, landing ground proximity sensor
essay evaluator	accuracy as compared to humans, feedback consistency	essay submissions, language rules, subject KB, grading rubrics	Score assignment, feedback generation, error highlighting	Text i/p check, grammar & plagiarism check, vocabulary analyzer

basic reason	accuracy in identifying threats, minimize false alarms, timely response	lab perimeters, personnel, firing mechanism, alarm system	cameras, motion detectors, heat sensors, facial recognition system, scanners	Guns, turret movements, communication systems, various safety
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Q5] Categorize a shopping bot for an offline store of book.
 Ans

- a) Partially Observable → can't observe all aspects of environment simultaneously. Has limited info about customer preferences, inventory changes by human staff.
- b) Stochastic → Has unpredictable elements. Customer behaviour is unpredictable, book availability changes over time, etc. Outcomes are not predicted precisely due to random factors.
- c) Sequential → Bot's decision affect future state & options. Eg :- Current recommendations will affect subsequent customer satisfaction.
- d) Dynamic → Environment changes while bot is operating & doesn't wait for bot to act. Inventory changes, staff moves around, preference may shift during interaction.
- e) Discrete → Bot's action can be represented in distinct states. Book locations, customer queries, inventory items etc are countable entities.
- f) Multiagent → The environment contains multiple agents, which interact with each other & affect the environment simultaneously.

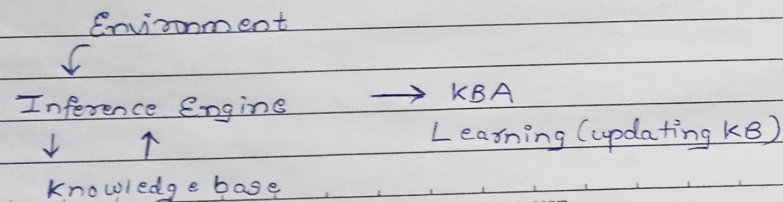
Q6] Model based

Utility based

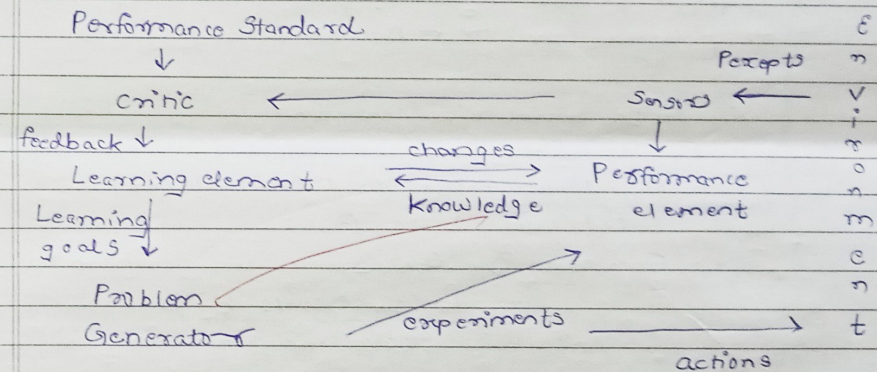
- | | |
|--|--|
| <p>① Maintains internal model of world, how world evolves independently of agent</p> <p>② Uses model to track the current state & predict the future possible states</p> <p>③ Based on model, it will select goals state achieving steps</p> <p>④ Keeps a track of the current state of world internally</p> <p>⑤ Less complex</p> <p>⑥ Eg:- Chess playing agent</p> | <p>① Not only maintains a model of the world but also assigns a utility value to/p</p> <p>② Evaluates different possible sequences & proceeds with one having max utility score</p> <p>③ Maximizes its utility function rather than a binary goal state</p> <p>④ Can handle preferences between goal & conflicting goal</p> <p>⑤ More complex</p> <p>⑥ Eg:- navigation agent finding optimal route</p> |
|--|--|

Q7] Explain architecture of knowledge based & an intelligent agent

Ans



- ① A KBA (Knowledge based agent) uses knowledge stored in Knowledge base (KB) to make decisions
- ② KB stores facts & rules i.e. information
- ③ An inference engine applies reasoning to derive conclusions
- ④ Perception collects data from environment
- ⑤ Actuators executes action
- ⑥ Learning module updates KB overtime
- Eg:- A medical diagnosis system



- ① A learning agent improves its performance overtime by learning from past experiences
- ② Learning element learns patterns from past actions while performance elements makes decisions based on learned data
- ③ Critic evaluates agent performance & gives a feedbacks.
- ④ The problem generator suggests new experience for learning.

99] Convert the following to predicate:-

① Anita travels by car if available otherwise travels by bus

$\Rightarrow \text{available}(\text{car}) \rightarrow \text{travels}(\text{Anita}, \text{car})$

$\Rightarrow \text{available}(\text{car}) \rightarrow \text{travels}(\text{Anita}, \text{Bus})$

② Bus route includes Andheri & Goregaon

$\Rightarrow \text{Route}(\text{Bus}, \text{Andheri}) \wedge \text{Route}(\text{Bus}, \text{Goregaon})$

③ car has a puncture & is unavailable

$\rightarrow \text{puncture}(\text{car}) \rightarrow \neg \text{Available}(\text{car})$

Will Anita travel via Goregaon?

Ans Via Forward Reasoning

From ③, $\text{Puncture}(\text{car}) \rightarrow \neg \text{Available}(\text{car})$

$\therefore \text{travel}(\text{Anita}, \text{Bus})$

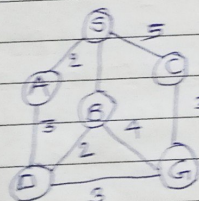
Since from rule ②, Bus travels via Goregaon &

Anita travels by bus

$\therefore$ Anita will travel via Goregaon

$\text{travel}(\text{Anita}, \text{Goregaon})$

10] BFS ($S \rightarrow G$) :-



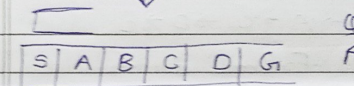
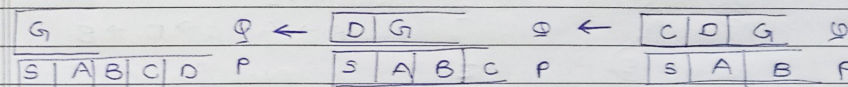
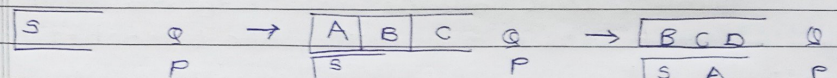
Ans. we will use a queue to solve Best First Search

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① make node S visited & enqueue it S

② Dequeue S & explore its neighbours i.e. A & B
maintain two queues (Q) & (P) for its neighbours

↑
a stack



③ Since Q is empty, pop elements from the stack / adjoint list,

Shortest path is $S \rightarrow B \rightarrow G$

& other include $S \rightarrow C \rightarrow G$

$S \rightarrow B \rightarrow D \rightarrow G$

$S \rightarrow A \rightarrow D \rightarrow G$

$S \rightarrow \{A, B, C\}$

$A \rightarrow \{E, D\}$

$B \rightarrow \{D, G\}$

$C \rightarrow \{G\}$

$D \rightarrow \{G\}$

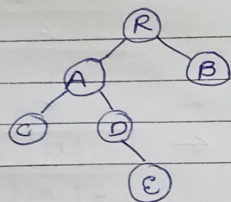
11] Explain DLS & IDS with example

Ans ① DLS is a variant of DFS that limits the depth of search, preventing it from exploring infinite or very deep search spaces

② It explores nodes to a specified depth before backtracking, useful for scenarios where the solⁿ is expected to be within a certain limit / depth

- ③ We begin at root node & define a maximum depth for the search; then recursively explore each node's neighbour, going deeper until the depth limit is reached or the goal is found; If it finds the goal, we return it or else backtrack & explore other branches

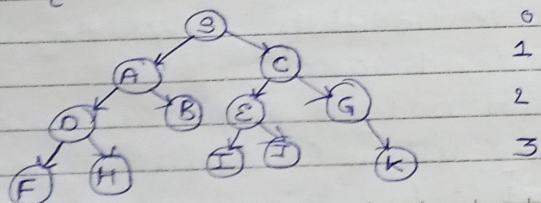
Eg:-



0
1 Limit(d) = 2
find: E
2 R → A → C → D → B
3 Not found E
now try: Limit(d) = 3
R → A → C → D → E
Found E

- ① DFS is search algo used in AI that blends the completeness of BFS with space efficiency of DFS.
- ② It explores graph progressively increasing the depth limit with each iteration, until goal is found.
- ③ Begin with root node & perform DFS with a depth limit 1; after each iteration perform DFS with depth limit incremented by 1.
- ④ continue until goal state is found.

iteration → 1st {S} (d=0)
2nd {S → A → C} (d=1)
3rd {S → A → D → B → C → E → G} (d=2)



Q12]

Explain Hill Climbing & its drawbacks. State the limitations with steepest ascent.

Ans

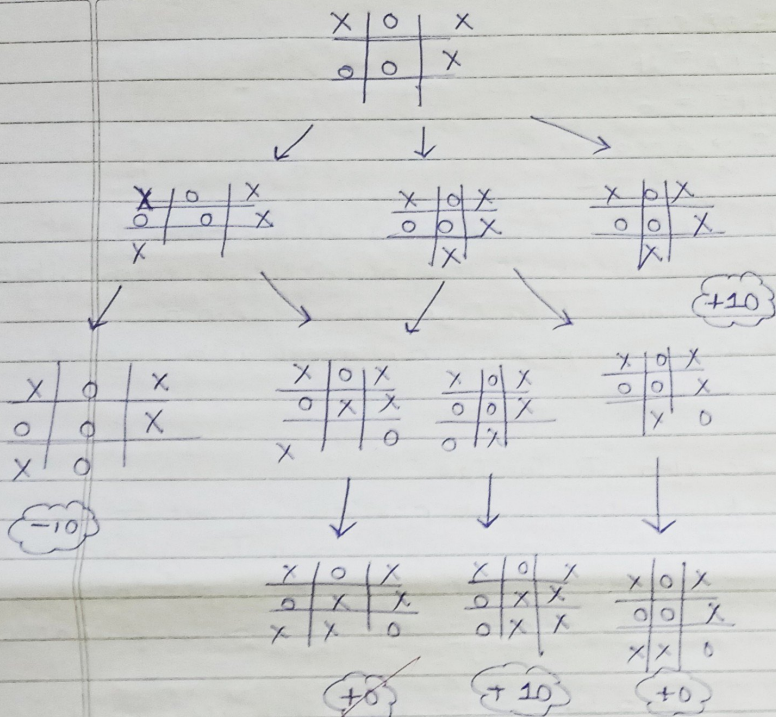
- ① It starts with an initial random solution; generating neighbouring solutions by applying small changes (mutations) to current solution.
 - ② A heuristic function is used to evaluate each of neighbouring sol's quality; we compare the quality & update if its better ^{repeat these steps}.
 - ③ we end until termination condition is met (max of iteration, satisfactory sol's quality, etc).
 - ④ It gets stuck in local optima; it's a better sol' but not than global optima.
 - ⑤ It can converge prematurely to suboptimal sol' if initial sol' isn't good.
 - ⑥ It can converge slowly to optimal sol's especially for problems with many local optima.
 - ⑦ While, steepest ascent hill climbing makes a greedy choice by selecting best neighbouring sol' at each step.
 - ⑧ It also doesn't explore sol's space effectively.
 - ⑨ Thus, it's sensitive to initial sol' state leading to poor convergence.
- Sol'n :- Stochastic Hill climbing {randomness}
stimulated Annealing

Q13]

Simulated Annealing:-

Ans

- ① It's a probabilistic optimization technique inspired by annealing process in metallurgy. And it is used to find global optimum of function with multiple local optima.



Q16] Explain $\alpha\beta$ pruning

Ans ① Adversarial search are used in scenarios where agents have conflicting goals, Eg:- Tic-tac-toe where player's gain is other's loss. The primary goal is to determine optimal strategy for player

② $\alpha\beta$ pruning is an optimization technique where it reduces the no. of nodes evaluated in game tree by eliminating branches that can't influence the final decision

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③ It maintains α , a maximizer; guarantee best/high value & β , a minimizer; guarantee best/lowest value.

④ As the algo traverses the tree; it updates the value. If finds a move worse than current α for maximizer or β for minimizer, it prunes/cuts off the branch.

$$\alpha = \max(\alpha, \text{child})$$

If $\alpha \geq \beta$ then prune the remaining children

$$\beta = \min(\beta, \text{child})$$

⑤ Thus it reduces complexity & enhances performance

Q17]

$$\begin{array}{r} 5 \quad 4 \quad 3 \quad 2 \quad 1 \\ + \quad 5 \quad 8 \quad N \quad 0 \\ m \quad \sigma \quad R \quad E \\ m \quad \sigma \quad n \quad e \quad y \\ c_3 \quad c_2 \quad c_1 \end{array}$$

① $m=1$ ($\because$ carry over is possible from sum of 2 single digit no in col 4)

② $\therefore 9 + m = \sigma$ (atleast 9)

$$\therefore 9 = 2/9 \quad \therefore 9 + m = 9/10$$

$$\therefore \sigma = 0/1 \quad \text{But } m=1 \quad \therefore \sigma = 0$$

③ If there's a carry $\therefore E=9$ & $N=0$

$$\text{But } 0=0 \quad \therefore \text{no carry \& } E=9 \& \sigma=0$$

④ If no carry in col 2 to 3 $\therefore E=N$ #impossible

$$\therefore \text{there's a carry \& } N+E \& \sigma=1$$

⑤ If a carry from col 1 to 2 $\therefore N+R = E \bmod 10$

$$\& N=E+1 \text{ so } E+1+R=E \bmod 10 \quad \therefore R=9 \text{ (but } 9=9)$$

$$\therefore \text{carry from col 1 to 2: } c_1=1 \& R=8$$

⑥ To produce carry $c_1=1$ $\therefore D+E = 10 + 7$ ($7 \neq 0 \& 1$)

$$\therefore D+E = \text{atleast } = 12 \text{ (0 is at most 9 \& E at least 5)}$$

$$\therefore N \text{ is atleast } 7 \& N=E+1 \text{ or } \Rightarrow E=5/6$$

⑦ If $E = 6$ & DTE atleast 12

$$\therefore D = 7$$

but $N = \varepsilon + 1$ & N would be also $\neq$ impossible

$$\therefore E=5 \quad \& \quad N=6$$

⑧ $0 + \epsilon$ is at least $t/2$

$$\therefore D = 7 \text{ ; } Y = 2$$

Hence,
$$\begin{array}{r} 9567 \\ + 1085 \\ \hline 10652 \end{array}$$

S $\rightarrow$ 9
E $\rightarrow$ 5
N $\rightarrow$ 6
D $\rightarrow$ 7
M $\rightarrow$ 1
O $\rightarrow$ 0
R $\rightarrow$ 8
Y $\rightarrow$ 2

Q18] PEAs of wumpus world. How it generates the percept Sequence

Ans The Wumpus world is classic AI environment used to demonstrate how an agent can reason & navigate a uncertain world with potential dangers.

Percepts :- ① Stench : presence of wumpus in adjacent room

④ Breeze : presence of pit in adjacent room

③ glitter: presence of gold in current room

④ Bump: Perceived when agent tries to move into wall

⑤ Scream: Heard when wumpus is killed with arrow

Environment :- ① Partially Observable: Perceives limited info based on percept above

② Deterministic as in effects of actions are predictable

③ Static $\rightarrow$ layout of world remains the same

④ Discrete → Agent moves in a specific locations on grid.
Actions :- move forward : move one step in current direction
turn left : rotate 90° left

turn left : rotate 90° left

turn right: rotate 90° right

shoot: fire an arrow in current direction to kill

Wumpus (limited to one arrow)

Goal: Navigate the cave; avoid falling into pits & being eaten by wumpus & find goal/while using as few actions as possible

Percept Sequence -

① stench :- If agent is in room adjacent to Wumpus, it will perceive stench

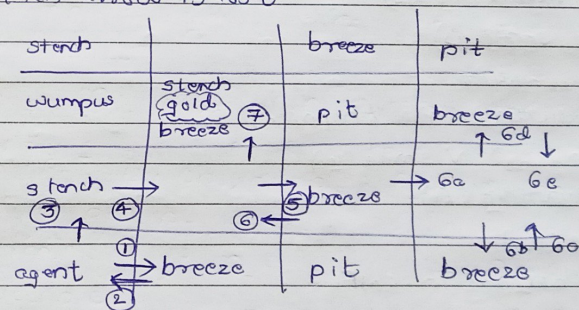
② Breeze :- If adjacent to pit; then breeze

③ Glitter :- If in room with gold, then glitter

④ No percept :- Not in the above

⑦ Agent can't directly sense its own location; so need a logical reasoning to locate its presence & dangers.

⑥ Wumpus can be only killed with single arrow, if a miss then arrow is lost



- Q19] a) All people who are graduating are happy
 b) All happy people smile
 c) Someone is graduating

d) Is someone smiling?

Ans 1) Convert to FOL :-

- a) $\forall x (\text{graduating}(x) \rightarrow \text{happy}(x))$
 b) $\forall x (\text{happy}(x) \rightarrow \text{smile}(x))$
 c) $\exists x \text{ graduating}(x)$

d) $\exists x \text{ smile}(x)$

2) Convert to CNF :-

Eliminate implication by $a \rightarrow b \equiv \neg a \vee b$

- a) $\forall x (\neg \text{graduating}(x) \vee \text{Happy}(x))$
 b) $\forall x (\neg \text{Happy}(x) \vee \text{smile}(x))$
 c) $\exists x \text{ graduating}(x)$
 d) $\neg \exists x \text{ smile}(x)$

Standardize variable apart

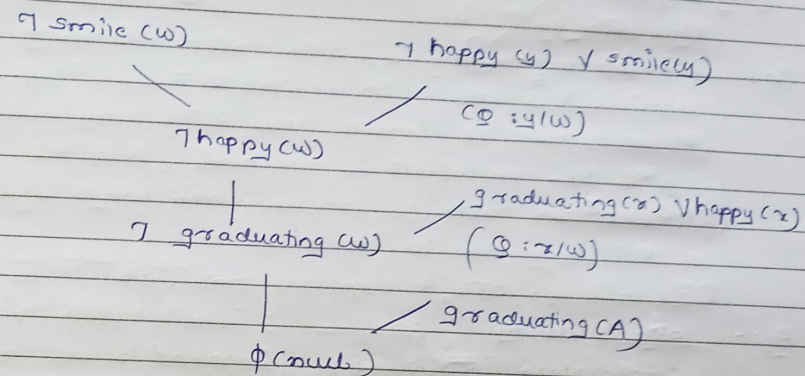
- $\forall x (\neg \text{graduating}(x) \vee \text{happy}(x))$
 $\forall y (\neg \text{happy}(y) \vee \text{smile}(y))$
 $\exists z \text{ graduating}(z)$
 $\neg \exists w \text{ smile}(w)$

drop universal quantifiers

- $\neg \text{graduating}(x) \vee \text{happy}(x)$
 $\neg \text{happy}(y) \vee \text{smile}(y)$
 $\text{graduating}(A)$
 $\neg \text{smile}(w)$

$\therefore$ all are in CNF

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 If fact F is to be proved then start with $\neg F$.
 It contradicts all other rules in KB
 Process stops when it returns Null clause



Hence, someone is smiling is proved

Q20] Explain Modus ponens

Ans It's a fundamental rule of logic that states that
 If a conditional statement ($P \rightarrow Q$) is true,
 & the antecedent (P) is also true, then
 consequent (Q) must be true

Eg:- P : It's raining $\rightarrow$ } P1
 Q : ground is wet. }
 it's raining } P2

$\therefore$ ground is wet (conclusion)
 if crow is a bird then fly crow is a bird
 $P \rightarrow Q$ P
 $\therefore$ crow can fly

Also, $P \rightarrow Q$, P is the representation.
 $\therefore Q$

Q11] Explain forward & backward reasoning chaining algo with eg

Ans: ① Forward chaining is a data driven reasoning technique & starts with available facts & applies inference rules to derive new facts, continuing a goal or conclusion is reached

② Moves from facts to conclusion & may lead to inefficiency due to irrelevant rules

③ It employs BFS, & explores multiple rules

Eg:- sky is cloudy (P) ; it will rain (r)
 temperature is low (q) ; ground is wet (w)

rule :- $(P \wedge q) \rightarrow r$

$\therefore (r \rightarrow w)$

$\therefore$ ground is wet.

A fire alarm system
 # that detects smokes & triggers alarm

④ Backward chaining begins with a goal & works backward to determine facts / rules

⑤ It's goal driven & is efficient facts.

⑥ Uses DFS & focus on one path

Eg:- a medical diagnosis that starts disease hypothesis & move backwards to identify symptoms

⑦ more efficient for targeted queries as it narrows search to relevant facts

goal :- Is ground wet?

we find :- If it rains, ground is wet

FOR EDUCATIONAL USE

we need to prove its raining
 $\therefore$ we find a rule, if ^{sky}cloud is cloudy, temp is low, then it will rain.

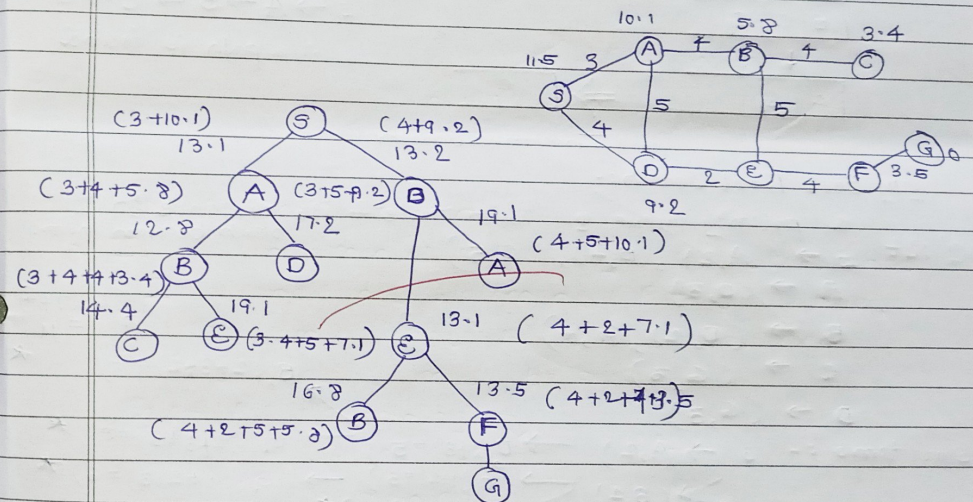
A fact found:- sky is cloudy & temp is low
 $\therefore$ it will rain

C: it rains $\therefore$ ground is wet.

By backward chaining

Corrected questions:-

A* :- eg $\rightarrow$

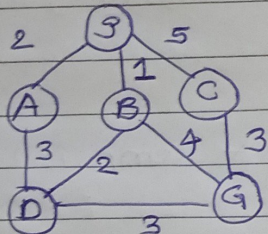


Uninformed Search

Q11]

Breadth First Search :-

Let's assume cost S to B as 1



queue(Q) :- push the neighbours of a node after marking it as visited
adj list

$$S = \{A, B, C\}$$

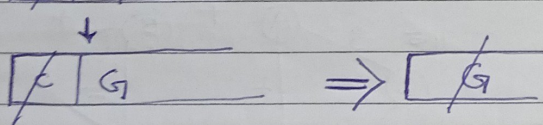
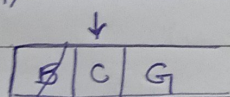
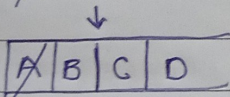
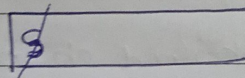
$$A = \{B, D, G\}$$

$$B = \{S, D, G\}$$

$$C = \{S, G\}$$

$$D = \{A, B, G\}$$

$$G = \{G, B, C\}$$



$$S \rightarrow A \rightarrow B \rightarrow C \rightarrow G$$

$\therefore$ a path is found

Informed Search

Q11] Best First Search :-

adj list a priority queue(Q) with node & its distance to reach between them.

$$S \rightarrow (A, 2), (B, 1), (C, 5)$$

$$A \rightarrow (D, 3)$$

$$B \rightarrow (D, 2), (G, 4)$$

$$C \rightarrow (G, 3)$$

$$D \rightarrow (G, 3)$$

$$\text{Start from S :- } S \rightarrow (A, 2) \rightarrow (D, (2+3)=5) \rightarrow (G, (2+3+3)=8) \star$$

$$\rightarrow (B, 1) \rightarrow (D, (1+2)=3) \rightarrow (G, (1+2+3)=6) \star$$

$$\rightarrow (C, 5) \rightarrow (G, (1+4)=5) \star$$

$$\rightarrow (G, (5+3)=8) \star$$

$\min(11, 8, 6, 5, 8)$ is shortest path
 $\therefore$ Shortest path is found

'S $\rightarrow$ B $\rightarrow$ G' with cost heuristic 5